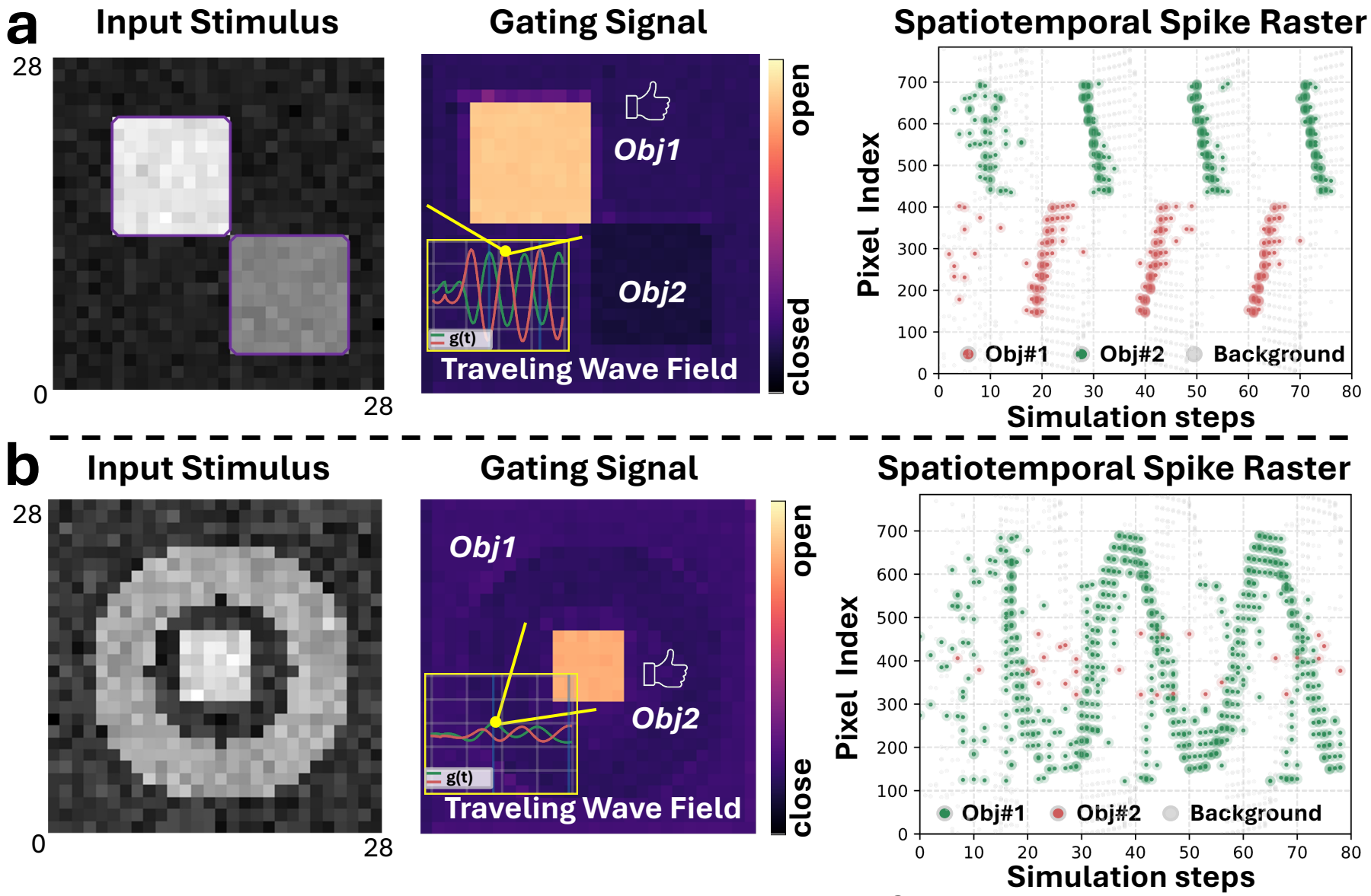


Figure 3. **Proof-of-concept validation of the S^2-Net across two distinct simulated tasks. (a)** Disjoint Object Discovery. This experiment demonstrates object binding via rhythmic gating and spatiotemporal wave fields. Rhythmic gating signals reproduce attention rhythms (e.g., Theta oscillations), where bright ("Open") and dark ("Closed") phases selectively bind features. Leveraging graph-structured delays, the framework generates traveling waves to maintain distinct temporal representations for disjoint objects. **(b)** Nested Topological Binding. This experiment addresses the more complex binding of topologically nested objects. Similar to the disjoint task, rhythmic gating and spatiotemporal spike raster collaborate to preserve distinct temporal representations despite severe spatial entanglement. Note: The sparser red dots corresponding to Obj #2 simply reflect its smaller physical size relative to the ring (i.e., Obj #1).